

DQueZ Carr

MCY 612

Dr. Wei Hoa

Cloud Security Assignment 5

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MCY 612 Cloud Security Assignment 5

In lab 5, you have learned how to evaluate an ELB system via its access log file. In MCY 611 cloud computing course, you have learned CloudFront and completed a CloudFront lab. In this assignment, you will read CloudFront logging

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/logging.html>

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/AccessLogs.html>

and then implement CloudFront standard logs.

Requirements

You need to use Cloudfront with logging enabled to accelerate a static website on S3. You can use lab 6 of MCY 611 as an example to implement this assignment. In lab 6, we did not enable CloudFront logging. In Step 1 of Lab 6, you need to create an additional S3 bucket specifically for storing CloudFront log files. The bucket name must follow this format: **your_nku_username**-logs-bucket. You need to modify enabling.json at step 5 of lab 6 to enable logging. The logging section of enabling.json in lab 6 is as follow:

```
"Logging": {
  "Enabled": false,
  "IncludeCookies": true,
  "Bucket": "",
  "Prefix": ""
},
```

Please look at

<https://awscli.amazonaws.com/v2/documentation/api/latest/reference/cloudfront/create-distribution.html>

to understand the meaning of each field in the logging section. Then modify enabling.json to enable logging.

You need to implement this assignment with AWS CLI.

Report

Introduction

The goal of Lab 5 in MCY 611 Cloud Computing was to assess an Elastic Load Balancer (ELB) system by examining its access log files in order to gain a better understanding of network performance and traffic. The current assignment builds on this foundation by examining and analyzing CloudFront logging to learn more about the distribution and accessibility of content across AWS's content delivery network. Finding important fields in the CloudFront logs, comprehending user behavior, and spotting any irregularities or trends in the delivery of material are the goals. Students had to obtain CloudFront log files from an S3 bucket, look at fields like x-edge-location, cs-uri-stem, and sc-status, and assess their meaning in relation to security and performance monitoring in order to finish this experiment.

Materials

AWS Cloudshell terminal

AWS CLI

AWS Version

aws-cli/2.23.13 Python/3.12.6 Linux/6.1.127-135.201.amzn2023.x86_64 exec-env/CloudShell
exe/x86_64.amzn.2023

AWS Configure

AWS Access Key ID [*****FLZO2]

AWS Secret Access Key [*****23fZK]

Default region name [us-east-1]

Default output format [json]
Security Group sg-*****602b
VPC vpc-*****6175
Subnet ID subnet-*****6ede

4. Methods

Make a bucket via the following mb command:

```
~ $ aws s3 mb s3://carrd12
make_bucket: carrd12
```

Delete public access block

Set the bucket policy with this command:

```
~ $ aws s3api put-bucket-policy --bucket carrd12 --policy file://policy.json
~ $
```

Download our MSCYS program webpage via the wget command, which downloads files from the Internet.

```
wget https://onlinedegrees.nku.edu/programs/technology/ms-cybersecurity/
```

An index.html file should be saved under your current directory.

Copy index.html to your bucket via the following command

```
~ $ aws s3 cp index.html s3://carrd12/index.html
upload: ./index.html to s3://carrd12/index.html
~ $
```

Create a distribution configuration json file named enabling.json in the current folder. It defines a CloudFront distribution.

```
    "Logging": {
      "Enabled": false,
      "IncludeCookies": true,
      "Bucket": "",
      "Prefix": ""
    },
    "ID": "G1", "ID": "G1", "ID": "G1"
```

Create the distribution via the following command:

```
~ $ aws cloudfront create-distribution --distribution-config file://enabling.json
{
  "Location": "https://cloudfront.amazonaws.com/2020-05-31/distribution/E2J3F5PNJ9P97C",
  "ETag": "E3AMGG4RQ1JQEV",
  "Distribution": {
    "Id": "E2J3F5PNJ9P97C",
    "ARN": "arn:aws:cloudfront::881490118823:distribution/E2J3F5PNJ9P97C",
    "Status": "InProgress",
    "LastModifiedTime": "2025-04-21T01:39:34.244000+00:00",
    "InProgressInvalidationBatches": 0,
    "DomainName": "d1a020g9qdb3c3.cloudfront.net",
    "ActiveTrustedSigners": {
      "Enabled": false,
      "Quantity": 0
    },
    "ActiveTrustedKeyGroups": {
      "Enabled": false,
      "Quantity": 0
    },
    "DistributionConfig": {
      "CallerReference": "carrd12-distribution",
...skipping...
...skipping...
  }
  "Location": "https://cloudfront.amazonaws.com/2020-05-31/distribution/E2J3F5PNJ9P97C",
  "ETag": "E3AMGG4RQ1JQEV",
  "Distribution": {
    "Id": "E2J3F5PNJ9P97C",
    "ARN": "arn:aws:cloudfront::881490118823:distribution/E2J3F5PNJ9P97C",
```

```
    "Status": "InProgress",
    "LastModifiedTime": "2025-04-21T01:39:34.244000+00:00",
    "InProgressInvalidationBatches": 0,
    "DomainName": "d1a020g9qdb3c3.cloudfront.net",
    "ActiveTrustedSigners": {
      "Enabled": false,
      "Quantity": 0
    },
    "ActiveTrustedKeyGroups": {
      "Enabled": false,
      "Quantity": 0
    },
    "DistributionConfig": {
      "CallerReference": "carrd12-distribution",
      ...skipping...
    },
    "Location": "https://cloudfront.amazonaws.com/2020-05-31/distribution/E2J3F5PNJ9P97C",
    "ETag": "E3AMGG4RQ1JJQEV",
    "Distribution": {
      "Id": "E2J3F5PNJ9P97C",
      "ARN": "arn:aws:cloudfront::881490118823:distribution/E2J3F5PNJ9P97C",
      "Status": "InProgress",
      "LastModifiedTime": "2025-04-21T01:39:34.244000+00:00",
      "InProgressInvalidationBatches": 0,
      "DomainName": "d1a020g9qdb3c3.cloudfront.net",
      "ActiveTrustedSigners": {
        "Enabled": false,
        "Quantity": 0
      }
    }
  },
  ...skipping...
}
```

AWS CloudShell

```
    "ActiveTrustedKeyGroups": {
      "Enabled": false,
      "Quantity": 0
    },
    "DistributionConfig": {
      "CallerReference": "carrd12-distribution",
      "Aliases": {
: ...skipping...
{
  "Location": "https://cloudfront.amazonaws.com/2020-05-31/distribution/E2J3F5PNJ9P97C",
  "ETag": "E3AMGG4RQ1JQEV",
  "Distribution": {
    "Id": "E2J3F5PNJ9P97C",
    "ARN": "arn:aws:cloudfront::881490118823:distribution/E2J3F5PNJ9P97C",
    "Status": "InProgress",
    "LastModifiedTime": "2025-04-21T01:39:34.244000+00:00",
    "InProgressInvalidationBatches": 0,
    "DomainName": "d1a020g9qdb3c3.cloudfront.net",
    "ActiveTrustedSigners": {
      "Enabled": false,
      "Quantity": 0
    },
    "ActiveTrustedKeyGroups": {
      "Enabled": false,
      "Quantity": 0
    },
    "DistributionConfig": {
      "CallerReference": "carrd12-distribution",
      "Aliases": {
```

AWS CloudShell

```

        "Quantity": 0
      },
      "...skipping..."
    ],
    "...skipping..."
  ],
  {
    "Location": "https://cloudfront.amazonaws.com/2020-05-31/distribution/E2J3F5PNJ9P97C",
    "ETag": "E3AMGG4RQ1JQEV",
    "Distribution": {
      "Id": "E2J3F5PNJ9P97C",
      "ARN": "arn:aws:cloudfront::881490118823:distribution/E2J3F5PNJ9P97C",
      "Status": "InProgress",
      "LastModifiedTime": "2025-04-21T01:39:34.244000+00:00",
      "InProgressInvalidationBatches": 0,
      "DomainName": "d1a020g9qdb3c3.cloudfront.net",
      "ActiveTrustedSigners": {
        "Enabled": false,
        "Quantity": 0
      },
      "ActiveTrustedKeyGroups": {
        "Enabled": false,
        "Quantity": 0
      },
      "DistributionConfig": {
        "CallerReference": "carrd12-distribution",
        "Aliases": {
          "Quantity": 0
        },
        "DefaultRootObject": "",
        "...skipping..."
      }
    }
  }
]

```

AWS Clo

```

"Location": "https://cloudfront.amazonaws.com/2020-05-31/distribution/E2J3F5PNJ9P97C",
"ETag": "E3AMGG4RQ1JQEV",
"Distribution": {
  "Id": "E2J3F5PNJ9P97C",
  "ARN": "arn:aws:cloudfront::881490118823:distribution/E2J3F5PNJ9P97C",
  "Status": "InProgress",
  "LastModifiedTime": "2025-04-21T01:39:34.244000+00:00",
  "InProgressInvalidationBatches": 0,
  "DomainName": "d1a020g9qdb3c3.cloudfront.net",
  "ActiveTrustedSigners": {
    "Enabled": false,
    "Quantity": 0
  },
  "ActiveTrustedKeyGroups": {
    "Enabled": false,
    "Quantity": 0
  },
  "DistributionConfig": {
    "CallerReference": "carrd12-distribution",
    "Aliases": {
      "Quantity": 0
    },
    "DefaultRootObject": "",
    "Origins": {
...skipping...
{
  "Location": "https://cloudfront.amazonaws.com/2020-05-31/distribution/E2J3F5PNJ9P97C",
  "ETag": "E3AMGG4RQ1JQEV",
  "Distribution": {

```

AWS Cloud


```
"Id": "E2J3F5PNJ9P97C",
"ARN": "arn:aws:cloudfront::881490118823:distribution/E2J3F5PNJ9P97C",
"Status": "InProgress",
"LastModifiedTime": "2025-04-21T01:39:34.244000+00:00",
"InProgressInvalidationBatches": 0,
"DomainName": "d1a020g9qdb3c3.cloudfront.net",
"ActiveTrustedSigners": {
  "Enabled": false,
  "Quantity": 0
},
"ActiveTrustedKeyGroups": {
  "Enabled": false,
  "Quantity": 0
},
"DistributionConfig": {
  "CallerReference": "carrd12-distribution",
  "Aliases": {
    "Quantity": 0
  },
  "DefaultRootObject": "",
  "Origins": {
    "Quantity": 1,
    :...skipping...
  }
  "Location": "https://cloudfront.amazonaws.com/2020-05-31/distribution/E2J3F5PNJ9P97C",
  "ETag": "E3AMGG4RQ1JQEV",
  "Distribution": {
    "Id": "E2J3F5PNJ9P97C",
    "ARN": "arn:aws:cloudfront::881490118823:distribution/E2J3F5PNJ9P97C",
```

```
ARN: "arn:aws:cloudfront:us-east-1:861156118825:distribution/E25513FN5513FC",
  "Status": "InProgress",
  "LastModifiedTime": "2025-04-21T01:39:34.244000+00:00",
  "InProgressInvalidationBatches": 0,
  "DomainName": "d1a020g9qdb3c3.cloudfront.net",
  "ActiveTrustedSigners": {
    "Enabled": false,
    "Quantity": 0
  },
  "ActiveTrustedKeyGroups": {
    "Enabled": false,
    "Quantity": 0
  },
  "DistributionConfig": {
    "CallerReference": "carrd12-distribution",
    "Aliases": {
      "Quantity": 0
    },
    "DefaultRootObject": "",
    "Origins": {
      "Quantity": 1,
      "Items": [
```

```
"Origins": {
  "Quantity": 1,
  "Items": [
    {
      "Id": "my-origin",
      "DomainName": "carrd12.s3.amazonaws.com",
      "OriginPath": "",
      "CustomHeaders": {
        "Quantity": 0
      },
      "S3OriginConfig": {
        "OriginAccessIdentity": ""
      },
      "ConnectionAttempts": 3,
      "ConnectionTimeout": 10,
      "OriginShield": {
        "Enabled": false
      },
      "OriginAccessControlId": ""
    }
  ]
},
"OriginGroups": {
  "Quantity": 0
},
"DefaultCacheBehavior": {
  "TargetOriginId": "my-origin",
  "TrustedSigners": {
    "Enabled": false.
```

```
        "Quantity": 0
    },
    "TrustedKeyGroups": {
        "Enabled": false,
        "Quantity": 0
    },
    "ViewerProtocolPolicy": "allow-all",
    "AllowedMethods": {
        "Quantity": 2,
        "Items": [
            "HEAD",
            "GET"
        ],
        "CachedMethods": {
            "Quantity": 2,
            "Items": [
                "HEAD",
                "GET"
            ]
        }
    },
    "SmoothStreaming": false,
    "Compress": false,
    "LambdaFunctionAssociations": {
        "Quantity": 0
    },
    "FunctionAssociations": {
        "Quantity": 0
    }
```

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```
        "Quantity": 0
    },
    "FieldLevelEncryptionId": "",
    "GrpcConfig": {
        "Enabled": false
    },
    "ForwardedValues": {
        "QueryString": true,
        "Cookies": {
            "Forward": "none"
        },
        "Headers": {
            "Quantity": 0
        },
        "QueryStringCacheKeys": {
            "Quantity": 0
        }
    },
    "MinTTL": 3600,
    "DefaultTTL": 86400,
    "MaxTTL": 31536000
},
"CacheBehaviors": {
    "Quantity": 0
},
"CustomErrorResponses": {
    "Quantity": 0
},
```



```
},
"Comment": "",
"Logging": {
  "Enabled": false,
  "IncludeCookies": true,
  "Bucket": "",
  "Prefix": ""
},
"PriceClass": "PriceClass_All",
"Enabled": true,
"ViewerCertificate": {
  "CloudFrontDefaultCertificate": true,
  "SSLSupportMethod": "vip",
  "MinimumProtocolVersion": "TLSv1",
  "CertificateSource": "cloudfront"
},
"Restrictions": {
  "GeoRestriction": {
    "RestrictionType": "none",
    "Quantity": 0
  }
},
"WebACLId": "",
"HttpVersion": "http2",
"IsIPV6Enabled": true,
"ContinuousDeploymentPolicyId": "",
"Staging": false
}
```

:

```
    "Logging": {
      "Enabled": false,
      "IncludeCookies": true,
      "Bucket": "",
      "Prefix": ""
    },
    "PriceClass": "PriceClass_All",
    "Enabled": true,
    "ViewerCertificate": {
      "CloudFrontDefaultCertificate": true,
      "SSLSupportMethod": "vip",
      "MinimumProtocolVersion": "TLSv1",
      "CertificateSource": "cloudfront"
    },
    "Restrictions": {
      "GeoRestriction": {
        "RestrictionType": "none",
        "Quantity": 0
      }
    },
    "WebACLId": "",
    "HttpVersion": "http2",
    "IsIPV6Enabled": true,
    "ContinuousDeploymentPolicyId": "",
    "Staging": false
  }
}
(CEND)
```

Check the information about a distribution via the following command:

```
~ $ aws cloudfront get-distribution --id E2J3F5PNJ9P97C
{
  "ETag": "E3AMGG4RQ1JQEV",
  "Distribution": {
    "Id": "E2J3F5PNJ9P97C",
    "ARN": "arn:aws:cloudfront::881490118823:distribution/E2J3F5PNJ9P97C",
    "Status": "Deployed",
    "LastModifiedTime": "2025-04-21T01:39:34.244000+00:00",
    "InProgressInvalidationBatches": 0,
    "DomainName": "d1a020g9qdb3c3.cloudfront.net",
    "ActiveTrustedSigners": {
      "Enabled": false,
      "Quantity": 0
    },
    "ActiveTrustedKeyGroups": {
      "Enabled": false,
      "Quantity": 0
    },
    "DistributionConfig": {
      "CallerReference": "carrd12-distribution",
      "Aliases": {
        "Quantity": 0
      },
      "DefaultRootObject": "",
      "Origins": {
        "Quantity": 1,
        "Items": [
          {

```

AWS CloudShell

:■


```
        {
            "Id": "my-origin",
            "DomainName": "carrd12.s3.amazonaws.com",
            "OriginPath": "",
            "CustomHeaders": {
                "Quantity": 0
            },
            "S3OriginConfig": {
                "OriginAccessIdentity": ""
            },
            "ConnectionAttempts": 3,
            "ConnectionTimeout": 10,
            "OriginShield": {
                "Enabled": false
            },
            "OriginAccessControlId": ""
        }
    ],
},
"OriginGroups": {
    "Quantity": 0
},
"DefaultCacheBehavior": {
    "TargetOriginId": "my-origin",
    "TrustedSigners": {
        "Enabled": false,
        "Quantity": 0
    },
},
```

```
    },
    "TrustedKeyGroups": {
      "Enabled": false,
      "Quantity": 0
    },
    "ViewerProtocolPolicy": "allow-all",
    "AllowedMethods": {
      "Quantity": 2,
      "Items": [
        "HEAD",
        "GET"
      ],
      "CachedMethods": {
        "Quantity": 2,
        "Items": [
          "HEAD",
          "GET"
        ]
      }
    },
    "SmoothStreaming": false,
    "Compress": false,
    "LambdaFunctionAssociations": {
      "Quantity": 0
    },
    "FunctionAssociations": {
      "Quantity": 0
    },
  },
```

: █

```
    },
    "FieldLevelEncryptionId": "",
    "GrpcConfig": {
      "Enabled": false
    },
    "ForwardedValues": {
      "QueryString": true,
      "Cookies": {
        "Forward": "none"
      },
      "Headers": {
        "Quantity": 0
      },
      "QueryStringCacheKeys": {
        "Quantity": 0
      }
    },
    "MinTTL": 3600,
    "DefaultTTL": 86400,
    "MaxTTL": 31536000
  },
  "CacheBehaviors": {
    "Quantity": 0
  },
  "CustomErrorResponses": {
    "Quantity": 0
  },
  "Comment": "",
```

```
"Comment": "",
"Logging": {
  "Enabled": false,
  "IncludeCookies": true,
  "Bucket": "",
  "Prefix": ""
},
"PriceClass": "PriceClass_All",
"Enabled": true,
"ViewerCertificate": {
  "CloudFrontDefaultCertificate": true,
  "SSLSupportMethod": "vip",
  "MinimumProtocolVersion": "TLSv1",
  "CertificateSource": "cloudfront"
},
"Restrictions": {
  "GeoRestriction": {
    "RestrictionType": "none",
    "Quantity": 0
  }
},
"WebACLId": "",
"HttpVersion": "http2",
"IsIPV6Enabled": true,
"ContinuousDeploymentPolicyId": "",
"Staging": false
}
}
```

```

    "Logging": {
      "Enabled": false,
      "IncludeCookies": true,
      "Bucket": "",
      "Prefix": ""
    },
    "PriceClass": "PriceClass_All",
    "Enabled": true,
    "ViewerCertificate": {
      "CloudFrontDefaultCertificate": true,
      "SSLSupportMethod": "vip",
      "MinimumProtocolVersion": "TLSv1",
      "CertificateSource": "cloudfront"
    },
    "Restrictions": {
      "GeoRestriction": {
        "RestrictionType": "none",
        "Quantity": 0
      }
    },
    "WebACLId": "",
    "HttpVersion": "http2",
    "IsIPV6Enabled": true,
    "ContinuousDeploymentPolicyId": "",
    "Staging": false
  }
}
}
}
(END)

```

When the status turns to “Deployed”, open a browser and visit your CloudFront URL

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5. Testing

For step 8 of lab 6, you only need to test the CloudFront URL from one location by using the Pingdom site. You need to repeat the CloudFront URL test twice. After the first test, you need to clear browser cache (history) before performing the second test.

Where is your Pingdom testing location? What are the latencies of the two tests?

URL	Test From	Load Time
https://d1a020g9qdb3c3.cloudfront.net/index.html	Tokyo, Japan	1.34s
https://d1a020g9qdb3c3.cloudfront.net/index.html	Tokyo, Japan	1.28s

Retrieve the CloudFront log file from the s3 bucket. If the log file is compressed in .gz format, you need to use `gzip -d` to decompress the log file <https://www.geeksforgeeks.org/gzip-command-linux/>. Show the corresponding CloudFront log entries for the two tests in your report.

Use the log entries to explain why the latencies of the two tests are different?

Use the log entries to explain the location of the edge server where the Pingdom requests are served. For CloudFront edge locations, please look at <https://www.feitsui.com/en/article/3>

Conclusion

In conclusion, this lab expanded on the information acquired from ELB logging in previous labs by giving practical experience in getting and analyzing CloudFront access logs. I was able to assess the efficacy of content distribution and spot possible problems by deciphering log entries, figuring out important metrics, and comprehending the structure and significance of different fields. This exercise illustrated how AWS services like CloudFront and S3 can be utilized in collaboration to offer scalable, secure, and effective content delivery. It also reaffirmed the significance of logging in cloud settings for monitoring and security purposes.

References

Cloudfront and edge function logging - amazon Cloudfront. (n.d.-d).

<https://docs.aws.amazon.com/AmazonCloudFront/latest/DeveloperGuide/logging.html>

Create-distribution¶. create-distribution - AWS CLI 2.26.5 Command Reference. (n.d.).

<https://awscli.amazonaws.com/v2/documentation/api/latest/reference/cloudfront/create-distribution.html>